A Correlation with sequence length

Figure 1 shows the distribution of user history lengths and their corresponding normalized DC components, indicating that our method for calculating the frequency of the user history is not strongly correlated with the history length. The

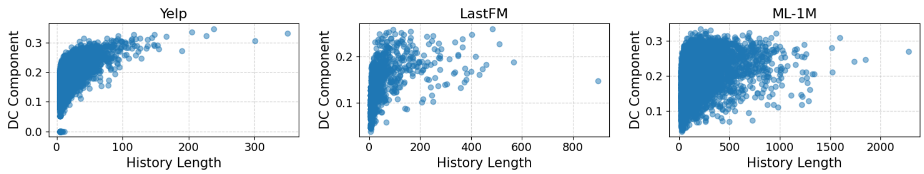

Fig. 1: Normalized DC component compared to the user history length.

weak correlation of DC frequency component and history length is also shown in Table 1, indicating the Pearson-r correlation and p value.

Dataset	**Pearson-r**	**p-value**
Yelp	0.589	0.000
LastFM	0.485	0.000
ML-1M	0.321	0.000

Table 1: The Pearson-r correlation, and p value when comparing the normalized DC frequency component with the history length.

B Changing encoded sequence length

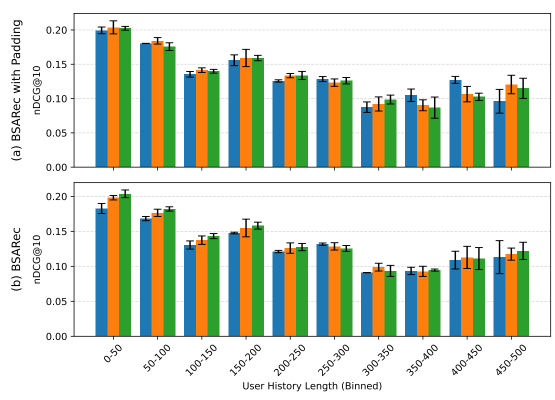

Fig. 2: Ranking performance of BSARec on ML-1M with reflect padding and zero padding.

Figure 2 compares reflect padding and zero padding when considering the length of the *user histories*, which are the actual recorded user-item interaction sequences rather than encoded to a specific length. For all three encoded sequence lengths ℓ, a similar trend over user history lengths is present for both types of padding. There are no notable differences between encoded sequence length ℓ due to the large standard deviations.

C Full tables

Table 2: Performance with different padding methods for BSARec. The best scores are marked bold. The column `Diff.` denotes the relative difference between zero padding and the best remaining padding method. Statistically significant differences $p < .05$ are identified using a t-test and marked with *.

Dataset	Metric	Zero	Cyclic	Reflect	Symmetric	Diff.
LastFM	HR@5	0.0497	0.0479	**0.0499**	0.0490	-0.37
	HR@10	0.0686	0.0730	**0.0760**	0.0701	-9.66*
	HR@20	0.1017	0.1073	**0.1114**	0.1072	-8.73*
	NDCG@5	**0.0347**	0.0329	0.0345	0.0346	0.42
	NDCG@10	0.0408	0.0411	**0.0430**	0.0414	-5.15
	NDCG@20	0.0490	0.0497	**0.0518**	0.0507	-5.47*
ML-1M	HR@5	0.1892	0.1946	**0.1947**	0.1945	-2.82
	HR@10	0.2784	0.2807	**0.2817**	0.2801	-1.14
	HR@20	0.3832	0.3864	**0.3882**	0.3860	-1.27
	NDCG@5	0.1271	**0.1317**	0.1315	0.1306	-3.44
	NDCG@10	0.1558	0.1594	**0.1595**	0.1582	-2.28
	NDCG@20	0.1822	0.1860	**0.1863**	0.1849	-2.18
Yelp	HR@5	0.0178	0.0187	**0.0193**	0.0190	-7.43*
	HR@10	0.0310	0.0323	**0.0332**	0.0328	-6.58*
	HR@20	0.0513	0.0548	**0.0556**	0.0554	-7.77*
	NDCG@5	0.0112	0.0117	**0.0121**	0.0119	-7.79*
	NDCG@10	0.0154	0.0161	**0.0166**	0.0164	-7.22*
	NDCG@20	0.0205	0.0217	**0.0222**	0.0220	-7.77*

Table 3: Performance comparison of BSARec with the Fourier transform (F) and the wavelet transform (W). The best average score is bolded. `Diff` shows improvement in percentage. *significant at $p < .05$.

	Beauty			**LastFM**			**ML-1M**		
Metric	F	W	Diff (%)	F	W	Diff (%)	F	W	Diff (%)
HR@5	**0.0508**	0.0508	0.05	**0.0477**	0.0440	8.33	**0.1888**	0.1852	1.95*
HR@10	**0.0712**	0.0705	0.86	**0.0703**	0.0642	9.43*	**0.2746**	0.2736	0.36
HR@20	**0.0987**	0.0978	0.92	**0.1015**	0.0989	2.60	**0.3814**	0.3773	1.10
NDCG@5	**0.0368**	0.0363	1.45	**0.0332**	0.0311	6.77	**0.1263**	0.1244	1.55
NDCG@10	**0.0433**	0.0426	1.62*	**0.0405**	0.0376	7.79	**0.1539**	0.1528	0.72
NDCG@20	**0.0503**	0.0495	1.58	**0.0483**	0.0462	4.63	**0.1808**	0.1790	1.01

Table 4: Results on seven Sequential recommendation (SR) benchmarks. The best and second-best scores are bold and underlined, respectively. The column `Diff.` denotes the relative difference between BSARec and the best baseline model. Statistically significant differences $p < .05$ are identified using a t-test and marked with *.

Dataset	**Metric**	**SASRec**	**BERT4Rec**	**FMLPRec**	**DuoRec**	**FEARec**	**BSARec**	**Diff.**
Beauty	HR@5	0.0297	0.0341	0.0347	<u>0.0517</u>	**0.0518**	0.0508	-1.87*
	HR@10	0.0466	0.0537	0.0535	0.0705	**0.0712**	<u>0.0712</u>	-0.13
	HR@20	0.0703	0.0802	0.0793	0.0961	<u>0.0971</u>	**0.0987**	1.62
	NDCG@5	0.0191	0.0221	0.0227	<u>0.0372</u>	**0.0372**	0.0368	-1.23
	NDCG@10	0.0246	0.0284	0.0288	0.0432	**0.0435**	<u>0.0433</u>	-0.40
	NDCG@20	0.0305	0.0351	0.0353	0.0497	<u>0.0500</u>	**0.0503**	0.49
LastFM	HR@5	0.0362	0.0294	0.0393	<u>0.0407</u>	0.0400	**0.0477**	17.12*
	HR@10	0.0546	0.0495	<u>0.0591</u>	0.0547	0.0543	**0.0703**	18.94*
	HR@20	0.0837	0.0754	<u>0.0839</u>	0.0829	0.0785	**0.1015**	21.01*
	NDCG@5	0.0244	0.0191	0.0253	<u>0.0294</u>	0.0285	**0.0332**	12.99
	NDCG@10	0.0303	0.0256	0.0316	<u>0.0338</u>	0.0331	**0.0405**	19.73*
	NDCG@20	0.0376	0.0320	0.0379	<u>0.0409</u>	0.0392	**0.0483**	18.07*
ML-1M	HR@5	0.1263	0.1474	0.1302	<u>0.1851</u>	0.1814	**0.1888**	2.02
	HR@10	0.2008	0.2234	0.2090	0.2678	<u>0.2700</u>	**0.2746**	1.73
	HR@20	0.3052	0.3263	0.3187	0.3731	<u>0.3737</u>	**0.3814**	2.05*
	NDCG@5	0.0815	0.0974	0.0832	<u>0.1245</u>	0.1218	**0.1263**	1.45
	NDCG@10	0.1054	0.1220	0.1085	<u>0.1512</u>	0.1503	**0.1539**	1.80*
	NDCG@20	0.1316	0.1479	0.1361	<u>0.1777</u>	0.1764	**0.1808**	1.74*
Sports and Outdoors	HR@5	0.0137	0.0164	0.0163	**0.0243**	<u>0.0238</u>	0.0237	-2.40
	HR@10	0.0221	0.0275	0.0256	**0.0351**	0.0346	<u>0.0349</u>	-0.54
	HR@20	0.0350	0.0432	0.0397	<u>0.0502</u>	0.0500	**0.0504**	0.35
	NDCG@5	0.0087	0.0106	0.0106	**0.0171**	<u>0.0168</u>	0.0164	-4.07*
	NDCG@10	0.0114	0.0141	0.0136	**0.0206**	<u>0.0203</u>	0.0200	-2.77*
	NDCG@20	0.0146	0.0181	0.0171	**0.0244**	<u>0.0242</u>	0.0239	-1.95*
Toys and Games	HR@5	0.0340	0.0283	0.0378	<u>0.0481</u>	0.0473	**0.0502**	4.35*
	HR@10	0.0476	0.0429	0.0531	<u>0.0650</u>	0.0642	**0.0672**	3.38*
	HR@20	0.0654	0.0654	0.0729	<u>0.0881</u>	0.0878	**0.0917**	4.03*
	NDCG@5	0.0231	0.0193	0.0257	<u>0.0358</u>	0.0352	**0.0369**	3.10*
	NDCG@10	0.0275	0.0240	0.0307	<u>0.0412</u>	0.0406	**0.0424**	2.77*
	NDCG@20	0.0320	0.0296	0.0357	<u>0.0471</u>	0.0466	**0.0485**	3.15*
Yelp	HR@5	0.0122	0.0190	0.0152	**0.0203**	<u>0.0192</u>	0.0182	-10.38*
	HR@10	0.0212	<u>0.0334</u>	0.0259	**0.0340**	0.0330	0.0309	-9.31*
	HR@20	0.0357	0.0559	0.0438	**0.0569**	<u>0.0561</u>	0.0518	-8.96*
	NDCG@5	0.0076	<u>0.0118</u>	0.0095	**0.0127**	0.0118	0.0114	-10.85*
	NDCG@10	0.0105	<u>0.0164</u>	0.0129	**0.0172**	0.0162	0.0154	-10.08*
	NDCG@20	0.0142	<u>0.0221</u>	0.0174	**0.0229**	0.0220	0.0207	-9.65*
MIND	HR@5	0.0690	**0.1143**	0.0624	0.1121	<u>0.1141</u>	0.1087	-4.87
	HR@10	0.1068	**0.1739**	0.0987	0.1692	<u>0.1710</u>	0.1633	-6.11
	HR@20	0.1586	**0.2506**	0.1529	0.2429	<u>0.2456</u>	0.2354	-6.08
	NDCG@5	0.0462	<u>0.0755</u>	0.0412	0.0743	**0.0760**	0.0724	-4.74
	NDCG@10	0.0584	**0.0947**	0.0529	0.0926	<u>0.0943</u>	0.0899	-5.07
	NDCG@20	0.0714	**0.1141**	0.0665	0.1112	<u>0.1131</u>	0.1081	-5.24